Progress Report

- Increment 2 -

Group: RuneScape Web App

1. **Team Members**

Miles Burkart (mjb23b) Github: 7Limes

Colton Bregoff (clb23g) Github: cbregoffid

William Kilduff (wrk24) Github: Killiam-duff

Agustin Camus (ac21bi) Github: C-Agus

Ricardo Furst (rcf22b) Github: ricfrst

2. **Project Title and Description**

Project Title: Grand Exchange Strategy Backtester

Project Description: A web application that lets Old School RuneScape players test trading strategies against real historical Grand Exchange data. The user defines a trading rule (conditions for buying and selling based on price, margin, and volume), and the application reports how that strategy would have performed. Existing tools tell players what to buy right now; RuneReport answers "would this strategy have worked?"

3. **Accomplishments and overall project status during this increment**

During this increment, we completed most of the core RuneReport functionality, including the Home page, Browse page, Individual Item page, navigation, item searching, filtering, and sorting. We also implemented the ability to create and save placeholder reports using browser Local Storage and view them through the saved_reports page. The project is progressing well, with most planned features completed. The remaining major tasks are implementing the Strategy Backtester and developing the actual report generation and analysis functionality

4. **Challenges, changes in the plan and scope of the project and things that went wrong during this increment**

As opposed to last increment where we were only dealing with the skeleton of the website and placeholders, this time we implemented real data and metrics fetched live from the OSRS API to provide up to date statistics to the users. While there were no mistakes/errors, it proved to be more of a challenge than simply dealing with dummy data.

As far as changes in the scope of the project go, that was a little more challenging. Bigger features like the strategy backtester and the item reports were pushed back to increment 3 as our final big features that really sell our website. You can see more in the functional requirements document, but features that were pushed back to increment 3 include:

- The backtester
- Individual item reports
- Live search results in home page
- Saving individual reports
- More low priority polishing.

The individual reports and backtester are the high priority items, as main/core features that set the website apart.

5. Team Member Contribution for this increment

1. *the **progress report***

   Miles Burkart: N/A

   Colton Bregoff: Question 6 and 7

   William Kilduff: Question 3

   Agustin Camus: Question 2

   Ricardo Furst: Question 4

2. *the **requirements and design document***

   Miles Burkart: N/A

   Colton Bregoff: Question 2

   William Kilduff: Question 1, Question 4, Question 5, and part of Question 7

   Agustin Camus: Part of Question 7

   Ricardo Furst: Questions 3 and 6

3. *the **implementation and testing document***

   Miles Burkart:

   Colton Bregoff: N/A

   William Kilduff: Question 3

   Agustin Camus: Questions 1 and 2

   Ricardo Furst:

4. *the **source code***

   Miles Burkart:

   Colton Bregoff: Best deals algorithm/panel on homepage and save report functionality in browser's local storage

   William Kilduff: Search on homepage/browsepage and placeholder reports

   Agustin Camus: Browse page sorting by price, daily volume and high alchemy value in both directions; buy and sell condition builder with AND/OR in Report Settings; working Saved Reports pagination; Saved Reports tab title fix; homepage search link fix.

   Ricardo Furst: browse page items and search, browse page page-scrolling functionality, mini placeholder reports on home page,

5. *the **video or presentation***

   Miles Burkart:

   Colton Bregoff: best deals panel and saved reports functionality

   William Kilduff:

   Agustin Camus:

   Ricardo Furst: conclusion and editing

6. Plans for the next increment

If this report if for the first or second increment, describe what are you planning to achieve in the next increment.

For the next increment, we plan to create and implement the backtesting algorithm for our app. In Increment 2, we laid the groundwork for the backtester by creating input fields on the browse page for buy and sell conditions, which can be combined with AND/OR clauses. The backtester will now use these conditions to simulate the strategy over a user-selected historical window, and the results will be shown on the report page based on the strategy's statistical performance (e.g., whether it turned a profit). In addition, we will make the "Save Report" button functional, so that it saves a snapshot of the report page to the browser's local storage. We would also like to add a few smaller features to make the final product feel more complete. We haven't decided on these yet, but we will discuss them at our first meeting of Increment 3. Above all, our main goal is to get the backtesting algorithm working in an effective and useful way.

7. Stakeholder Communication

RuneReport Project Update – Second Increment

Hello,

I wanted to provide a second progress update on the state of RuneReport, our web app designed to allow Old School RuneScape players to simulate their trading strategies against historical Grand Exchange prices and see how those strategies would have performed.

For this increment, we connected RuneReport to live Grand Exchange data. Players can now search, browse, and sort items, view current item information, and see the top five potential item deals on the homepage. We also added the ability to save reports, and built the strategy builder, where players can set the conditions for when the backtester should buy and sell an item.

We did hit a minor setback during this increment, though. The strategy builder ended up taking longer than planned because it depended on other features being finished first. Rather than rush the backtester, we decided to move it back to the next increment so it can get our full time and attention for the rest of the project.

For the third and final increment, we will build the backtester itself, which will run a player's strategy over a selected time window and show how it would have performed. Overall, the project is currently on track, and we look forward to sharing more with you in our next update.

Kind regards,
RuneReport Team

8. Link to video

https://drive.google.com/file/d/1nGFbc_GQWgjhGtlpNKXodIscLmAfW0ZE/view?usp=sharing